

Ziyu Xu – Full-text redline against original resume

Red-strikeout: ~~deleted~~ Blue underline: added

Ziyu_Xu_D1

Original baseline: CMU_Ziyu_original.tex. All visible text is compared in PDF reading order; moved content appears as deletion and insertion. Layout-only changes are described in changes.md.

Ziyu (Rick) Xu Robotics Institute, Carnegie Mellon University Research Interests: Robot Foundation Models, ~~Real-World Policy~~ VLA Adaptation, and ~~Learning-Augmented Control~~ Real-Robot Learning +1 7348005258 | rickx@andrew.cmu.edu

Education

Carnegie Mellon University (CMU) Aug. 2026 – Expected 2028 Master of Science in Robotics (MSR) Pittsburgh, PA
University of Michigan, Ann Arbor (UM) Oct. 2024 – May 2026 B.S.E. in Mechanical Engineering (Robotics Track);
Minor in Computer Science GPA: 3.85/4.0 (Top 5%) Core Coursework: Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization
Shanghai Jiao Tong University (SJTU) Sep. 2022 – Jul. 2026 B.S.E. in Electrical and Computer Engineering GPA: 3.56/4.0 (Top 20%) Core Coursework: Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems

Research Experiences and Internships

Unitree Robotics, R&D Department, Embodied AI Team May 2026 – Present Research Intern, Humanoid Robot Foundation Models Hangzhou, China ~~Closed-Loop Real-Robot Infrastructure for Humanoid Foundation-Model Policies: Policy Infrastructure:~~ Built and validated ~~an end- to-end policy infrastructure a closed-loop stack~~ on Unitree H2/G1 ~~humanoids; integrating teleoperation data collection; humanoids integrat- ing teleoperation,~~ VLA/WAM and whole-body policy adapters, rollout ~~logging and logging,~~ evaluation, controller interfaces, and policy ~~fine-tuning. Enabled launch with one command and closed-loop rollouts fine-tuning; enabled one-command launches~~ across ~~platforms; allowing policies and controllers to be integrated without modifying platform-specific launch workflows: platforms.~~ Embodiment VLA Adaptation & ~~and~~ Post-Training: ~~Conducted a structured survey of representative robot VLA; WAM; and physical-AI policies, especially work from NVIDIA GEAR Lab.~~ Deployed and fine-tuned Isaac GR00T/OpenPI VLA policies with SONIC ~~whole-body whole- body~~ control on Unitree ~~humanoids~~ using real-robot rollout traces. ~~Currently developing a Developing~~ phase-aware adaptation ~~method that aligns across embodiments by aligning~~ semantic task progress and execution ~~timing across embodiments for policy transfer and rollout-driven post-training: timing.~~ Reproducible ~~Real-World VLA Benchmark for Humanoid Manipulation: Evaluation:~~ Implemented an 8-task, 24-scene ~~manipulation benchmark with green-screen background replacement to evaluate foundation models under varied visual con- ditions: variation.~~ Built a PICO-based ~~scene-digitization scene digitization~~ and AR-guided reconstruction ~~pipeline that captured shelf-object layouts as with~~ reusable digital scene templates; and ~~registered target placements back into the physical workspace; layouts,~~ achieving 2.5 cm median placement error. Intelligent Control Lab, Carnegie Mellon University Aug. 2026 – Present Graduate Student Researcher, advised by Prof. Changliu Liu Pittsburgh, PA Whole-Body and Safe Locomotion Research (ongoing): Investigating whole-body control, safety with control barrier functions (CBFs), and world-model-based locomotion. Dexmate Bimanual Manipulation Demo, IROS 2026: Contributed to a bimanual block-stacking demonstra- tion combining agent planning with policy execution. Implemented the agent component, prepared the on-site environment, adapted the system to different grippers, and performed system calibration. Shanghai Artificial Intelligence Lab & IRMV, SJTU Apr. 2026 – Present Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang Shanghai, China ~~Context-Aware Re-encoding for~~ Iterative Visual KV Compression: ~~Identified representation Studied~~ errors ~~caused by from~~ reusing retained visual KV entries after ~~their spatial neighbors neighboring tokens~~ are ~~removed, and showed that these errors com- pound across repeated compression removed~~ in multi-turn and streaming inference. Developed a context-aware re-encoding ~~method that recomputes each retained set in its new context at the original positions; positions and benchmarked it against compared with~~ inheritance and restoration ~~methods~~ under matched memory and runtime budgets. Ziyu (Rick) Xu Research experience continued Computational Autonomy and Robotics Lab, UM Jun. 2025 – Apr. 2026 Research Assistant, Directed by Prof. Maani Ghaffari Ann Arbor, MI ~~Online Learning Lie-MPC for Robust ASV Control: Learning-Augmented Lie-MPC:~~ Developed a Neural-Fly-inspired learning-augmented Lie-MPC framework for and field-tested an autonomous surface ~~vehicle; estimating vehicle controller that esti- mates~~ residual wind/current disturbances online and ~~injecting learned corrects the~~ dynamics ~~corrections into the controller for robust trajectory tracking under hydrodynamic uncertainty. Implemented and evaluated the framework on a real ASV; achieved used by MPC. Achieved a 20% improvement im- provement~~ in trajectory-tracking performance over the baseline MPC controller during field experiments. ~~Field Perception for Aquatic Vegetation Tracking: Built a water-scene perception pipeline for autonomous sur- face vehicles; combining DINO/SAM2-based segmentation with Grounded-DINO and YOLO fine-tuning to detect submerged vegetation and~~

provide perception signals for navigation and autonomy experiments. ASV Hardware and Sensor Integration for Closed-Loop Field Autonomy: Designed sensor mounts and integrated GPS, lidar, IMU, and camera systems on an autonomous surface vehicle. Aligned and calibrated sensors to support field data collection and control in real-ASV experiments. Intelligent Robotics and Autonomy Lab, UM Oct. 2024 – Apr. 2026 Research Assistant, Directed by Prof. Vasileios Tzoumas Ann Arbor, MI Multi-Robot Active-SLAM Simulation & ROS2 Integration: and ROS2: Rebuilt a Unity + AirSim multi-drone simulation testbed with redesigned vehicle revised dynamics and sensing modules; then sensing; migrated SlideSLAM communication logics from ROS1 to ROS2 by refactoring topic interfaces and workflows for bandwidth-limited active-SLAM evaluation. Resource-Aware Multi-Robot Exploration Algorithms: Developed distributed exploration EXP3-style algorithms for bandwidth-limited robot teams, formulating communication-aware action selection with submodular objectives and bandit feedback in simulation to balance map coverage and communication cost. Surgical Computer Vision Detection, SJTU Apr. 2024 – Dec. 2024 Research Assistant, Directed by Prof. Yutong Ban Shanghai, China Foundation Segmentation Model Evaluation for Surgical Video: Evaluated Segment Anything Model variants on surgical video frames, analyzing domain shift and temporal mask stability around instruments, tissues, and tool-tissue interaction regions.

Publications and Manuscripts

Xiaohan Wang*, Ziyu Xu*, Xuyi Yang. “Re-encoding What You Keep After Every Compression: Retained Visual KV Is Computed in a Context That No Longer Exists”, Submitted to ICLR 2027. * Equal contribution. Xuyi Yang, Xiaohan Wang, Wenhao Zhang, and Ziyu Xu. “StreamJIT: Continuous Observation and Just-in-Time Deliberation for Proactive Streaming Video Assis- tants”, Assistants”, Submitted to ICLR 2027. Yinan Dong, Ziyu Xu, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari. “Online Learning-Enhanced Lie Algebraic Alge- braic MPC for Robust Autonomous Surface Vehicle Control”, IEEE Robotics and Automation Letters (RA-L), revised manuscript under review, 2025. C. Yuan, J. Jiang, K. Yang, Z. Xu, . . . , and Y. Ban. “Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis”, npj Digital Surgery, accepted, 2024. Selected

Awards

Gold Medal in 2023 Medal, University Physics Competition, by American Physical Society & American Astronomi- cal Society (top 1%) Third place in 2023 RoboMaster Campus Competition (top 10%) (2023); Dean’s Honor List, University of Michigan College of Engineering (2024, 2025) M Prize in 2024 Mathematical Contest in Modeling (top 13%) 2025).

Skills

Robot Learning & Control: Learning: VLA post-training, humanoid foundation models, real-robot deployment, whole- body policy, policy integration, RL, real- robot policy deployment, teleoperation/data collection, MPC, sim-to-real teleoperation, rollout evaluation Perception & Multimodal ML: ML and Systems: PyTorch, Python/C++, ROS/ROS2, Linux, Docker, Git, OpenCV, SAM2, Grounded-DINO, YOLO, DINO features, visual token compression, spatial grounding, LLMs fine-tuning Systems & Tools: ROS/ROS2, Python/C++, Unity + AirSim, Docker, Linux, Git, sensor integration, CAD, FEA, Manufacturing, embedded system KV compression